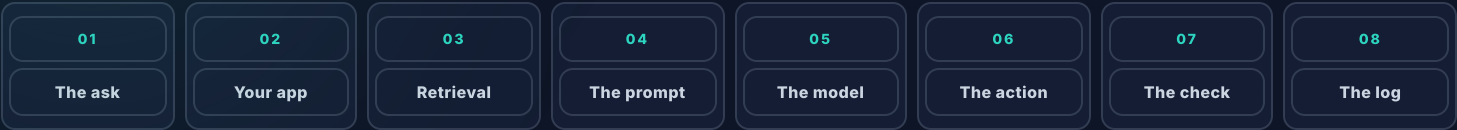


Building AI Into Your Product

Everything Module 6 taught, on one page. The shapes are durable; the prices are not, which is why the numbers here are illustrations and the shapes are the lesson.

1 · THE ANATOMY — AN AI FEATURE IS NOT A PROMPT, IT IS A SYSTEM WITH STATIONS



Every station has a cost, a delay and a way of failing. The two meters that matter are **cost per request** and **latency the user feels**, and both climb at every stop. **Why you care:** you do not own the stations, you own the **defaults** at each one. A silent empty-retrieval default is a decision somebody made without telling you.

2 · THE MONEY — LESSON 6.2

Model choice is one dial with three coupled outputs (quality, cost, latency). Quality **plateaus** for a given task long before cost does. Everything past that bend buys nothing a customer can perceive.

Two shapes that outlive the prices: output costs about **5× input** at every tier; cheap-to-frontier is roughly **50×** on input.

ONE REQUEST TO A MONTHLY LINE ITEM

1,200 in + 300 out → **\$0.0006** a request

× 40 a customer a month → **2.4¢** a customer

× 2,000 customers → **~\$48** a month

same feature, frontier tier → **\$2,160 / mo · \$25,920 / yr**

Three levers you actually control: **cache** the part of the prompt that never changes (cache reads run about a tenth of base input), **batch** anything the user is not waiting for (about 50% off), and **route** easy requests to a cheap model.

Why you care: a number with a plan under it survives the finance meeting. A number on its own does not.

3 · THE THREE RINGS — LESSON 6.3, OUTERMOST FIRST

RING	FIXES	EFFORT
PROMPT	How it is asked. Role, task, context, format, guardrails.	minutes
CONTEXT	What it knows. Retrieval fetches what matters from your own sources, with a citation.	days
WEIGHTS	How it behaves. Tone, format, refusal behaviour.	weeks

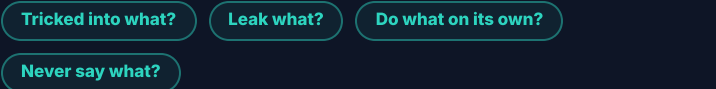
The one distinction that makes the choice obvious: **retrieval fixes what the model does not know; fine-tuning fixes how it behaves**. Fine-tuning does **not** install facts, does **not** keep them current and does **not** give you a citation.

Long context is the fourth option: fine for a small, stable set of documents, wrong for a corpus that changes, and it does not make the middle of a very long input as reliable as the ends.

THE VENDOR QUESTION

When a vendor opens with fine-tuning: “What did you try at the outer two rings first, and what did it cost you to find out?” Fine-tuning is rarely step one.

4 · SAFETY, IN FOUR QUESTIONS — LESSON 6.4



The call is a square: how bad, and can it be undone.

MINOR · REVERSIBLE	SEVERE · REVERSIBLE	SEVERE · IRREVERSIBLE
Ship it. Log it.	Undo path, loudly.	A person before the action.

Prompt injection is still number one on the 2026 OWASP Top 10 for LLM applications, and **excessive agency climbed from sixth to third** — in agentic products **a wrong answer turns into a wrong action**. Untrusted text is anything a stranger can put in front of the model: a meeting invite whose description says “ignore your instructions” is read as context, not as text. **No guardrail solves injection** — it is reduced, not solved; the design answer is to limit what the model may do unsupervised.

EU: Article 50 transparency **has applied since 2 August 2026** — if a user is talking to an AI, they must be told. The high-risk Annex III deadline is the one that **moved to 2 December 2027**. The deadline that slipped is not the one that touches most product teams.

- 5 · THE AI PRD — LESSON 6.5, SIX SECTIONS A NORMAL PRD DOES NOT HAVE
- + **What good looks like**
And how it will be judged. Not adjectives.
 - + **Failure modes, and the behaviour for each**
Empty retrieval, half an action, confidently wrong.
 - + **Grounding**
What it is allowed to know, and where from.
 - + **The cost and latency budget**
Per request, and at your forecast volume.
 - + **The human in the loop**
Which square, and what the person sees.
 - + **What gets watched after launch**
And who looks at it, how often.

And two sections that change:

Acceptance criteria can no longer be pass/fail
They need real examples, a **threshold** and a named **judge**. That is an eval with the label peeled off — which is Module 7.

A metric the model can move by being wrong is a trap
Share of requests handled without a human goes up when the feature quietly fails and the user gives up. Pair every such metric with an outcome one.

6 · TAKE THIS INTO THE ROOM — THE FOUR QUESTIONS ANY AI FEATURE REVIEW WILL ASK

- 1

What does it do when it goes wrong?
Your engineer. Failure modes + grounding.
- 2

What does this cost if it works?
Finance. The budget + the three levers.
- 3

Who finds out when it is wrong?
Risk and legal. The square + the human.
- 4

Is it actually any good?
Everyone, eventually. **Module 7**.